

C# 프로그래밍 — 6강 실습 퀴즈 정답지

동명대학교 게임학부
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1. 정답 1 — 주사위 대결 게임

핵심 포인트

- Player 클래스는 제공됨 — top-level 게임 로직만 작성
- 객체 배열 생성, foreach 순회, 최댓값 찾기
- 라운드별 승자 판정 후 점수 누적, 최종 우승자 출력

```
1 // top-level code
2 Player[] players = new Player[]
3 {
4     new Player("Player1"),
5     new Player("Player2"),
6     new Player("Computer")
7 };
8
9 Console.WriteLine("=== Dice Battle Game (5 rounds) ===\n");
10
11 for (int r = 1; r <= 5; r++)
12 {
13     Console.WriteLine("--- Round " + r + " ---");
14
15     int maxValue = 0;
16     string winner = "";
17
18     foreach (Player p in players)
19     {
20         int value = p.RollDice();
21         Console.WriteLine("    " + p.Name + " -> [" + value + "]"
22             );
23
24         if (value > maxValue)
25         {
26             maxValue = value;
27             winner = p.Name;
28         }
29     }
30
31     foreach (Player p in players)
32     {
33         if (p.Name == winner)
34             p.Score++;
```

```

34     }
35     Console.WriteLine(" * Winner: " + winner + "\n");
36 }
37
38 Console.WriteLine("=== Final Results ===");
39 Player best = players[0];
40 foreach (Player p in players)
41 {
42     Console.WriteLine(" " + p.Name + " : " + p.Score + " pts")
43     ;
44     if (p.Score > best.Score)
45         best = p;
46 }
47 Console.WriteLine("\n* Champion: " + best.Name
48     + " (" + best.Score + " pts)");
49
50 // --- class ---
51 class Player
52 {
53     public string Name;
54     public int Score;
55     private Random rand;
56
57     public Player(string name)
58     {
59         Name = name;
60         Score = 0;
61         rand = new Random();
62     }
63
64     public int RollDice()
65     {
66         return rand.Next(1, 7);
67     }
68 }

```

2. 정답 2 — 학생 성적 관리

핵심 포인트

- Student 클래스: 필드 4개, 생성자, 메서드 3개 (GetAverage, GetGrade, GetInfo)
- 평균 계산 시 3.0으로 나누어 소수점 결과를 얻음
- 1등/꼴등 찾기: students[0]으로 초기화 후 반복문에서 비교

```

1 // top-level code
2 Student[] students = new Student[]
3 {

```

```
4     new Student("Kim", 85, 92, 78),
5     new Student("Lee", 95, 88, 97),
6     new Student("Park", 72, 65, 80),
7     new Student("Choi", 60, 55, 48),
8     new Student("Jung", 88, 91, 85),
9 };
10
11 // Print all
12 Console.WriteLine("=== Grade Report ===\n");
13 foreach (Student s in students)
14 {
15     Console.WriteLine("    " + s.GetInfo());
16 }
17
18 // Find best, worst, class average
19 Student best = students[0];
20 Student worst = students[0];
21 double total = 0;
22
23 for (int i = 0; i < students.Length; i++)
24 {
25     double avg = students[i].GetAverage();
26     total += avg;
27
28     if (avg > best.GetAverage())
29         best = students[i];
30     if (avg < worst.GetAverage())
31         worst = students[i];
32 }
33
34 Console.WriteLine("\n=== Statistics ===\n");
35 Console.WriteLine("    * Top: " + best.Name
36     + " (Avg " + Math.Round(best.GetAverage(), 1) + ")");
37 Console.WriteLine("    * Bottom: " + worst.Name
38     + " (Avg " + Math.Round(worst.GetAverage(), 1) + ")");
39 Console.WriteLine("    * Class Avg: "
40     + Math.Round(total / students.Length, 1));
41
42
43 // --- class ---
44 class Student
45 {
46     public string Name;
47     public int Kor;
48     public int Eng;
49     public int Math;
50
51     public Student(string name, int kor, int eng, int math)
52     {
53         Name = name;
54         Kor = kor;
```

```

55         Eng = eng;
56         Math = math;
57     }
58
59     public double GetAverage()
60     {
61         return (Kor + Eng + Math) / 3.0;
62     }
63
64     public string GetGrade()
65     {
66         double avg = GetAverage();
67         if (avg >= 90) return "A";
68         if (avg >= 80) return "B";
69         if (avg >= 70) return "C";
70         if (avg >= 60) return "D";
71         return "F";
72     }
73
74     public string GetInfo()
75     {
76         return Name
77             + " | Kor:" + Kor + " Eng:" + Eng
78             + " Math:" + Math
79             + " | Avg: " + System.Math.Round(GetAverage(), 1)
80             + " | Grade: " + GetGrade();
81     }
82 }

```

3. 정답 3 — 가위바위보 게임

핵심 포인트

- 2개 클래스: Player (입력/랜덤), GameManager (진행/판정)
- top-level 코드는 단 2줄 — GameManager 패턴
- 승패 판정: 가위(0)→보(2), 바위(1)→가위(0), 보(2)→바위(1)
- 3판 2선승: while (user.Wins < 2 && computer.Wins < 2)

```

1  // top-level code
2  GameManager manager = new GameManager();
3  manager.Play();
4
5
6  // --- Player class ---
7  class Player
8  {
9      public string Name;
10     public int Wins;
11     private Random rand;

```

```
12
13     public Player(string name)
14     {
15         Name = name;
16         Wins = 0;
17         rand = new Random();
18     }
19
20     // Get hand from user input (0=scissors, 1=rock, 2=paper)
21     public int ChooseHand()
22     {
23         while (true)
24         {
25             Console.Write("    Choose (0=scissors, 1=rock, 2=
26                 paper): ");
27             string input = Console.ReadLine();
28
29             int value;
30             if (int.TryParse(input, out value)
31                 && value >= 0 && value <= 2)
32                 return value;
33
34             Console.WriteLine("    [Error] Choose 0, 1, or 2!");
35         }
36     }
37
38     // Computer random hand
39     public int RandomHand()
40     {
41         return rand.Next(0, 3);
42     }
43
44     // --- GameManager class ---
45     class GameManager
46     {
47         private Player user;
48         private Player computer;
49
50         public GameManager()
51         {
52             user = new Player("Player");
53             computer = new Player("Computer");
54         }
55
56         // Convert hand number to name
57         private string HandName(int hand)
58         {
59             if (hand == 0) return "scissors";
60             if (hand == 1) return "rock";
61             return "paper";
```

```

62     }
63
64     // Return: 0=draw, 1=user wins, 2=computer wins
65     private int Judge(int h1, int h2)
66     {
67         if (h1 == h2) return 0;
68
69         if ((h1 == 0 && h2 == 2) ||
70             (h1 == 1 && h2 == 0) ||
71             (h1 == 2 && h2 == 1))
72             return 1;
73
74         return 2;
75     }
76
77     public void Play()
78     {
79         Console.WriteLine("=== Rock-Paper-Scissors "
80             + " (Best of 3) ===\n");
81
82         int round = 0;
83         while (user.Wins < 2 && computer.Wins < 2)
84         {
85             round++;
86             Console.WriteLine("--- Round " + round + " ---");
87
88             int userHand = user.ChooseHand();
89             int compHand = computer.RandomHand();
90
91             Console.WriteLine("  " + user.Name + ": "
92                 + HandName(userHand)
93                 + " vs " + computer.Name + ": "
94                 + HandName(compHand));
95
96             int result = Judge(userHand, compHand);
97             if (result == 0)
98                 Console.WriteLine("  -> Draw!");
99             else if (result == 1)
100             {
101                 Console.WriteLine("    * "
102                     + user.Name + " wins!");
103                 user.Wins++;
104             }
105             else
106             {
107                 Console.WriteLine("    * "
108                     + computer.Name + " wins!");
109                 computer.Wins++;
110             }
111
112             Console.WriteLine("  [" + user.Name + " "

```

```

113         + user.Wins + "W / " + computer.Name
114         + " " + computer.Wins + "W]\n");
115     }
116
117     Console.WriteLine("=== Final Result ===");
118     if (user.Wins > computer.Wins)
119         Console.WriteLine("* " + user.Name
120             + " wins!");
121     else
122         Console.WriteLine("* " + computer.Name
123             + " wins!");
124 }
125 }

```

채점 기준 요약

항목	배점	세부 기준
클래스 설계	10	필드, 생성자, 메서드가 올바르게 구현되었는가
프로그램 로직	10	반복문, 조건문, 게임/통계 흐름이 정확한가
출력 결과	5	예시와 유사한 형식으로 깔끔하게 출력되는가
코드 스타일	5	top-level statements 사용, 클래스 하단 배치, 가독성
합계	30	